

Claude Managed Agents + Tracing

Practical AI PM cheat sheet from a controlled State Question Investigator exercise | Sept. 2026

Core product rule: AI interprets -> software enforces -> people authorize. The agent may investigate and suggest, but it does not approve Reviews, change Current State, or resolve Questions.

1. What the exercise was testing

- **Agent design:** Can a managed agent search project material deeply without crossing authority boundaries?
- **Tooling:** Can it work with a deliberately limited read/search toolset?
- **Tracing:** When an answer is wrong, can we tell whether the failure came from retrieval, missing context, or synthesis?
- **Iteration:** Can we change the instructions, rerun the same case, and verify the fix?

2. Agent setup

Job	Investigate one unresolved State Question and return an evidence packet plus a suggested answer for human review.
Allowed tools	Read, Grep, Glob.
Disabled	Write, Edit, Bash, Web Search, Web Fetch.
Sources	Mounted project files representing Current State, Reviews, Questions, Notes, Slack, and transcripts.
No confidence score	Evidence and provenance carry the weight instead of a number that could create automation bias.

3. Investigation order

- Check **Current State** for what is already established.
- Check related **open Reviews** and keep them clearly labeled as pending proposals.
- Check existing **Questions** to avoid duplicate work.
- Search raw sources deeply enough to find supporting, conflicting, and missing evidence.
- Preserve chronology. Newer evidence may qualify or supersede older evidence.
- Return a suggested answer, provenance, remaining unknowns, and the next human action.

4. What tracing is for

Evals tell you that the agent failed. Traces help you localize where the failure first became observable.	
Step	Question to ask
Retrieval	Did the agent find the right files or records?
Context	Did it receive enough relevant information, with the right chronology and provenance?
Interpretation	Did it correctly understand what the evidence establishes and what it does not?
Authority handling	Did it accidentally turn a proposal or pending Review into established fact?
Final output	Did the wording preserve uncertainty and keep the human decision explicit?

5. Controlled test set and results

Case	What it tested	Result
Q-101	Clear answer exists in a legal note.	PASS Minor wording / review-burden notes.
Q-102	Older working date vs newer evidence that timing is still TBD; related open Review exists.	PASS
Q-103	Ownership is proposed in an open Review but not yet established.	FAIL -> FIX -> PASS
Q-104	Answer is buried in an older transcript and absent from Current State.	PASS
Q-105	Evidence explicitly says ownership is unassigned; agent must not invent an owner.	PASS

6. The useful failure: Q-103

Observed failure: The agent found the right evidence and the correct open Review, but its first answer said "Priya Shah owns customer support training" and claimed the group had accepted that ownership.

Why that was wrong: The sources only showed that Priya offered to take ownership, Devon agreed to be backup, and an open Review proposed the assignment. Current State still said ownership was under review.

Trace checkpoint	Finding
Retrieval	Correct files found.
Source evidence	Correct and sufficient.
Review detection	Correctly recognized R-201 was open.
Synthesis	First wrong step. Pending proposal was upgraded to established fact.

7. Prompt fix that worked

- Open Reviews are **pending proposals**, never established facts.
- If a Question is covered by an open Review, use proposed/pending language unless Current State independently establishes the fact.
- Do not infer group approval from one person agreeing.
- If new evidence conflicts with Current State, say it **may support an update** rather than declaring Current State wrong or stale.

Retest: Same data + same Q-103 question -> agent correctly said ownership was not yet established and pointed the human to R-201.

8. Product takeaways

- **Read-only tools do not guarantee safe interpretation.** Authority mistakes can still happen in language.
- **Tracing is localization, not mind reading.** You may not see why the model made a leap, but you can still find the first observable wrong step.
- **Do not fix the prompt before diagnosing the failure.** Retrieval problems, context problems, and synthesis problems need different fixes.
- **Retest the exact same case after a fix.** Otherwise you cannot tell whether the change addressed the failure.
- **Watch review burden.** Correct agents can still create unnecessary follow-up work or over-cautious recommendations.

Portfolio-ready story: Built a read-only managed agent, designed controlled evaluation cases, found a genuine authority failure, used the trace to localize it to synthesis rather than retrieval, changed the instructions, and verified the fix with the same test case.

Exercise scope: Controlled learning exercise using a fake State dataset. Results are not production performance claims.